

Overall Reflection

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PHYS 250 Final Project

I learned a lot through this project, especially what it takes to produce a scientific paper alongside working code. I did a good job building an experiment largely on my own: a nine-qubit circuit, labeled events, training, holdout tests, and figures that tell one story from encoding to measurement. I built everything from scratch. This was also the first time I trained a model in a machine-learning sense. COBYLA tuning the rotation angles, comparing splits, and checking baselines gave me a clearer picture of how optimization and evaluation fit together, even though the setup is small. I am more comfortable now with the path from idea to methods to results to honest limitations.

The directional accuracy was weaker than I hoped, and several holdout readouts did not match the realized five-day labels. That was expected: strong market models often take years to develop and serious resources to maintain. Even so, it did not feel like a dead end. I built something amazing: the foundation of something that could possibly work. The model is definitely not perfect, and I know it could be better. At least I know what I would need to do next: more events, better labels, longer optimization, and richer features than five scores per event.

I found useful tools along the way, including AI-assisted coding and data work. For example, using AI to help scrape and organize web-sourced material was efficient when it worked, though it burned through a lot of tokens and I hit usage limits quickly. I still found that workflow helpful for moving faster on repetitive steps. I checked outputs against my notebook and simulators before trusting them in the paper.

Finance is not an area I have studied deeply. If I did this project again, I would not rely only on whether each asset was up or down over the five trading days after a headline. Events can trigger instant moves or delayed reactions when news reaches a wider audience. A single fixed window misses that. The model only predicts up or down per asset, not how large the move might be. I would like to explore a readout or classical post-processing step that speaks to magnitude, not just sign. I also see limits in treating USD like the other assets. The dollar index moves, but people are not “investing in USD” the way they hold equities or Bitcoin. Gold, the S&P 500, and crypto fit the story more naturally as risk-on or risk-off legs.

I had a great time on this project and I am very appreciative of the freedom to choose the topic, build the circuit, and learn from mistakes. I leave with a working prototype, a paper I can explain, and a short list of concrete improvements for whatever comes next.